

KKT conditions

Background material

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Basic notions

Consider the problem

$$\begin{aligned} \min_{x \in \mathcal{X}} f(x) \\ \text{subject to } g_i(x) \leq 0, \quad i = 1, \dots, m, \\ h_j(x) = 0, \quad j = 1, \dots, r, \end{aligned}$$

where $\mathcal{X} \subseteq \mathbb{R}^n$ is the **ground set** on which f , g and h are defined (often $\mathcal{X} = \mathbb{R}^n$; it may also hold the constraints we prefer not to dualize).

The **feasible set** is

$$\mathcal{F} = \{x \in \mathcal{X} \mid g_i(x) \leq 0, \quad i = 1, \dots, m, \quad h_j(x) = 0, \quad j = 1, \dots, r\}.$$

x^* is a global minimizer if $\forall x \in \mathcal{F}, f(x^*) \leq f(x)$.

x^* is a local minimizer if $\exists \epsilon > 0$ such that $\forall x \in \mathcal{B}(x^*, \epsilon) \cap \mathcal{F}, f(x^*) \leq f(x)$.

Lagrangian and Lagrangian dual function

We define the Lagrangian as

$$L(x, \lambda, \mu) = f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^r \mu_j h_j(x),$$

and the dual Lagrangian function

$$\mathcal{L}(\lambda, \mu) = \inf_{x \in \mathcal{X}} L(x, \lambda, \mu),$$

the infimum being taken over the ground set, and possibly equal to $-\infty$.

Lagrange multipliers: equality constraints

Consider the mathematical program

$$\begin{aligned} \min_{x \in \mathcal{X}} f(x) \\ \text{subject to } h_j(x) = 0, \quad j = 1, \dots, r, \end{aligned}$$

where $\mathcal{X} \subset \mathbb{R}^n$, $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $h_j : \mathbb{R}^n \rightarrow \mathbb{R}$, $j = 1, \dots, r$.

The **Lagrangian** of this problem is obtained by associating a Lagrange multiplier μ_j to each constraint function h_j :

$$L(x, \mu) = f(x) + \sum_{j=1}^r \mu_j h_j(x).$$

We can obtain very general conditions under which x^* is an optimal solution to the optimization problem, while only basic assumptions are made over $\mathcal{X}$ and the functions f and h_j , $j = 1, \dots, r$.

Optimality

Theorem

Assume that the Lagrangian associated to the problem

$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & f(x) \\ \text{s.t.} \quad & h_j(x) = 0, \quad j = 1, \dots, r, \end{aligned}$$

has a local minimizer $x^ \in \mathcal{X}$ when the multiplier vector μ is equal to μ^* . If $h_j(x^*) = 0, j = 1, \dots, r$, then x^* is a local minimizer of the constrained problem.*

Optimality

Proof.

Assume by contradiction that x^* is not a local minimizer of the constrained problem. Then $\forall \epsilon > 0$, $\exists \bar{x} \in \mathcal{X} \cap \mathcal{B}(x^*, \epsilon)$ such that $h_j(\bar{x}) = 0$, $j = 1, \dots, r$, and $f(\bar{x}) < f(x^*)$. Thus, $\forall \mu$,

$$\sum_{j=1}^r \mu_j h_j(x^*) = \sum_{j=1}^r \mu_j h_j(\bar{x}) = 0,$$

and

$$f(\bar{x}) + \sum_{j=1}^r \mu_j h_j(\bar{x}) < f(x^*) + \sum_{j=1}^r \mu_j h_j(x^*).$$

Taking $\mu = \mu^*$, the previous inequality contradicts that x^* is a local minimizer of the Lagrangian when $\mu = \mu^*$. □

Lagrange multipliers: inequality constraints

Same setting, with inequality constraints: $\mathcal{X} \subset \mathbb{R}^n$, $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$, $i = 1, \dots, m$.

Theorem

Assume that the Lagrangian associated to the problem

$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & f(x) \\ \text{s.t.} \quad & g_i(x) \leq 0, \quad i = 1, \dots, m, \end{aligned}$$

has a local minimizer $x^ \in \mathcal{X}$ when the multiplier vector λ is equal to λ^* . If $g_i(x^*) \leq 0$, $\lambda_i^* \geq 0$, and $\lambda_i^* g_i(x^*) = 0$, $i = 1, \dots, m$, then x^* is a local minimizer of the constrained problem.*

Lagrange multipliers: inequality constraints

Proof.

As previously, assume by contradiction that x^* is not a local minimizer of the constrained problem. Then $\forall \epsilon > 0$, $\exists \bar{x} \in \mathcal{X} \cap \mathcal{B}(x^*, \epsilon)$ such that $g_i(\bar{x}) \leq 0$, $i = 1, \dots, m$, and $f(\bar{x}) < f(x^*)$. Therefore, for $\lambda = \lambda^* \geq 0$,

$$\sum_{i=1}^m \lambda_i g_i(\bar{x}) \leq 0 \text{ and } \sum_{i=1}^m \lambda_i g_i(x^*) = 0.$$

Consequently,

$$f(\bar{x}) + \sum_{i=1}^m \lambda_i g_i(\bar{x}) < f(x^*) + \sum_{i=1}^m \lambda_i g_i(x^*),$$

contradicting that x^* is a local minimizer of the Lagrangian when $\lambda = \lambda^*$. □

Dual problem

The dual problem is

$$\begin{aligned} & \max_{\lambda \in \mathbb{R}^m, \mu \in \mathbb{R}^r} \mathcal{L}(\lambda, \mu) \\ & \text{such that } \lambda \geq 0. \end{aligned}$$

Important properties:

- The dual problem is always convex, i.e. $\mathcal{L}$ is always concave (even if the primal problem is not convex).
- The primal and dual (global) optimal values, f^* and $\mathcal{L}^*$, always satisfy the weak duality: $f^* \geq \mathcal{L}^*$.
- **Strong duality**: under some conditions (constraint qualifications), $f^* = \mathcal{L}^*$.

Saddle point

Since

$$\max_{\lambda \geq 0, \mu} L(x, \lambda, \mu) = \begin{cases} f(x) & \text{if } x \in \mathcal{F}, \\ +\infty & \text{otherwise,} \end{cases}$$

the primal problem reads $\min_{x \in \mathcal{X}} \max_{\lambda \geq 0, \mu} L(x, \lambda, \mu)$, and the dual problem $\max_{\lambda \geq 0, \mu} \min_{x \in \mathcal{X}} L(x, \lambda, \mu)$: weak duality is nothing but $\max \min \leq \min \max$.

Definition (Saddle point)

(x^*, λ^*, μ^*) , with $\lambda^* \geq 0$, is a saddle point of L if

$$L(x^*, \lambda, \mu) \leq L(x^*, \lambda^*, \mu^*) \leq L(x, \lambda^*, \mu^*) \quad \forall x \in \mathcal{X}, \forall \lambda \geq 0, \forall \mu.$$

(x^*, λ^*, μ^*) is a saddle point if and only if x^* is primal optimal, (λ^*, μ^*) is dual optimal, and the duality gap is null.

Working with the dual function

$\mathcal{L}$ is concave, being an infimum of functions that are affine in (λ, μ) , but it is usually **not differentiable**. A supergradient is however immediate.

Theorem

Let $\bar{x} \in \arg \min_{x \in \mathcal{X}} L(x, \lambda, \mu)$. Then, for all (λ', μ') ,

$$\mathcal{L}(\lambda', \mu') \leq \mathcal{L}(\lambda, \mu) + g(\bar{x})^T (\lambda' - \lambda) + h(\bar{x})^T (\mu' - \mu).$$

Proof.

$\mathcal{L}(\lambda', \mu') \leq L(\bar{x}, \lambda', \mu') = L(\bar{x}, \lambda, \mu) + g(\bar{x})^T (\lambda' - \lambda) + h(\bar{x})^T (\mu' - \mu)$, and $L(\bar{x}, \lambda, \mu) = \mathcal{L}(\lambda, \mu)$ by definition of $\bar{x}$. $\square$

The constraint values at the Lagrangian minimizer thus give a supergradient, the search direction of the **dual ascent** methods: $\lambda^{\nu+1} = \max\{\lambda^\nu + \alpha^\nu g(\bar{x}^\nu), 0\}$, the maximum projecting on $\lambda \geq 0$. As $\mathcal{L}$ is non-smooth, such a direction is not necessarily an ascent one: the step α^ν must be chosen accordingly.

Duality gap

Given a primal feasible solution x and a dual feasible solution (λ, μ) , the quantity $f(x) - \mathcal{L}(\lambda, \mu)$ is called the duality gap between x and (λ, μ) . Note that

$$f(x) - f^* \leq f(x) - \mathcal{L}(\lambda, \mu).$$

Therefore, if the duality gap is equal to 0, then x is primal-optimal (and similarly, λ and μ are dual-optimal).

From an algorithmic point of view, if strong duality holds, this provides a stopping criterion: if $f(x) - \mathcal{L}(\lambda, \mu) \leq \epsilon$, then $f(x) - f^* \leq \epsilon$.

Duality gap: local case

We can also define the dual Lagrangian function restricted to the ball $\mathcal{B}(x^*, \epsilon)$:

$$\mathcal{L}_{\mathcal{B}(x^*, \epsilon)}(\lambda, \mu) = \min_{x \in \mathcal{B}(x^*, \epsilon)} L(x, \lambda, \mu).$$

The weak duality still holds locally:

$$\mathcal{L}_{\mathcal{B}(x^*, \epsilon)}^* \leq f(x^*).$$

Under some conditions, the strong duality also holds:

$$\mathcal{L}_{\mathcal{B}(x^*, \epsilon)}^* = f(x^*).$$

Duality gap: local case

Note however that

$$\min_{x \in \mathbb{R}^n} L(x, \lambda, \mu) \leq \min_{x \in \mathcal{B}(x^*, \epsilon)} L(x, \lambda, \mu)$$

so

$$\mathcal{L}^* \leq \mathcal{L}_{\mathcal{B}(x^*, \epsilon)}^*.$$

Therefore, if x^* is a local minimizer and the strong duality locally holds,

$$\mathcal{L}^* \leq f(x^*).$$

The inequality can be strict.

Karush-Kuhn-Tucker (KKT) conditions

Consider $f, g_i, h_j \in C^1$, $i = 1, \dots, m$, $j = 1, \dots, r$, and the problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & g_i(x) \leq 0, \quad i = 1, \dots, m, \\ & h_j(x) = 0, \quad j = 1, \dots, r. \end{aligned}$$

Karush-Kuhn-Tucker (KKT) conditions:

$\nabla_x L(x, \lambda, \mu) = 0$	(stationarity)
$\lambda_i g_i(x) = 0$	(complementarity)
$g_i(x) \leq 0, \quad h_j(x) = 0 \quad \forall i, j$	(primal feasibility)
$\lambda_i \geq 0 \quad \forall i$	(dual feasibility)

Necessary conditions

Let x^* be a minimizer of f over $\mathcal{F} \cap \mathcal{B}(x^*, \epsilon)$, $\epsilon > 0$, and (λ^*, μ^*) be a dual solution if x is restricted to $\mathcal{B}(x^*, \epsilon)$, with a zero duality gap (the strong duality holds). Then

$$\begin{aligned} f(x^*) &= \mathcal{L}_{\mathcal{B}(x^*, \epsilon)}(\lambda^*, \mu^*) \\ &= \min_{x \in \mathcal{B}(x^*, \epsilon)} \left(f(x) + \sum_{i=1}^m \lambda_i^* g_i(x) + \sum_{j=1}^r \mu_j^* h_j(x) \right) \\ &\leq f(x^*) + \sum_{i=1}^m \lambda_i^* g_i(x^*) + \sum_{j=1}^r \mu_j^* h_j(x^*) \\ &\leq f(x^*) \end{aligned}$$

Thus, x^* is a minimizer of $L(x, \lambda^*, \mu^*)$ in $\mathcal{B}(x^*, \epsilon)$, and $\nabla_x L(x^*, \lambda^*, \mu^*) = 0$.

We have obtained the stationarity conditions.

Necessary conditions

The previous inequalities also imply $\sum_{i=1}^m \lambda_i^* g_i(x^*) = 0$ as $\sum_{i=1}^m \lambda_i^* g_i(x^*) \leq 0$, and consequently, $\lambda_i^* g_i(x^*) = 0, \forall i$.

This establishes the complementarity conditions.

If x^* is a global minimizer, we can replace $\mathcal{B}(x^*, \epsilon)$ by $\mathbb{R}^n$.

We can summarize our findings in the theorem below.

Theorem (KKT necessary conditions)

If $x^, (\lambda^*, \mu^*)$ are primal and dual solutions with a null duality gap, then $x^*, (\lambda^*, \mu^*)$ satisfy the KKT conditions.*

Strong duality

The strong duality assumption often plays a key role. How to ensure that it holds?

- **Linear programming**: it always holds, as soon as one of the two programs has a finite optimal value.
- **Convex programming**: the program is convex when $\mathcal{X}$ is convex, f and the g_i are convex, and the h_j are affine. The **Slater condition** is then sufficient: there exists a feasible x in the relative interior of $\mathcal{X}$ with $g_i(x) < 0$ for every non-affine g_i .
- **Nonconvex programming**: a constraint qualification is required. The most common, and the most demanding, is the linear independence constraint qualification (LICQ).

Nonconvex programming

Theorem (Necessary conditions)

If x^ is a local solution of*

$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & f(x) \\ \text{s.t.} \quad & g_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_j(x) = 0, \quad j = 1, \dots, r, \end{aligned}$$

where f , the g_i , $i = 1, \dots, m$, and the h_j , $j = 1, \dots, r$, are C^1 , and a constraint qualification condition holds at x^ , then $\exists(\lambda^*, \mu^*)$ such that the KKT conditions hold at (x^*, λ^*, μ^*) .*

Proof.

See Nocedal & Wright, "Numerical Optimization", Section 12.4. □

Sufficiency of KKT conditions

Let (x^*, λ^*, μ^*) satisfy the KKT conditions. Primal feasibility and complementarity give

$$L(x^*, \lambda^*, \mu^*) = f(x^*) + \sum_{i=1}^m \lambda_i^* g_i(x^*) + \sum_{j=1}^r \mu_j^* h_j(x^*) = f(x^*).$$

In the **convex case**, $L(\cdot, \lambda^*, \mu^*)$ is convex, so stationarity makes x^* a *global* minimizer of the Lagrangian over $\mathcal{X}$:

$$\mathcal{L}(\lambda^*, \mu^*) = L(x^*, \lambda^*, \mu^*) = f(x^*).$$

The duality gap is null (**strong duality**) and $x^*, (\lambda^*, \mu^*)$ are globally optimal for the primal and the dual, respectively.

In the **nonconvex case**, stationarity only makes x^* a critical point of $L(\cdot, \lambda^*, \mu^*)$: it may be a local minimizer, a saddle point, or even a maximizer, and $\mathcal{L}(\lambda^*, \mu^*) < f(x^*)$ remains possible. The KKT conditions are then necessary, under a constraint qualification, but not sufficient — not even locally, as the next slide shows.

KKT and the local duality gap

The necessary conditions were obtained from the **local** strong duality. Does the converse hold: do the KKT conditions restore a null gap on a small enough ball?

As $L(x^*, \lambda^*, \mu^*) = f(x^*)$, the local gap vanishes on $\mathcal{B}(x^*, \epsilon)$ **if and only if** x^* *minimizes* $L(\cdot, \lambda^*, \mu^*)$ on that ball. The KKT conditions only provide **stationarity**, which is strictly weaker. A constraint qualification cannot help: it is what makes the conditions *necessary* at a local solution, and says nothing about what they imply.

Counterexample

$\min_{x \in \mathbb{R}^2} x_1^2 - x_2^2$ subject to $h(x) = x_2 = 0$. The feasible set is the axis $x_2 = 0$, on which $f = x_1^2$: $x^* = (0, 0)$ is the unique global solution, and $L(x, \mu) = x_1^2 - x_2^2 + \mu x_2$ is stationary at x^* for $\mu^* = 0$. Regularity is not at fault: $\nabla h = (0, 1) \neq 0$, so **LICQ holds**, and on the critical cone $\{d \mid d_2 = 0\}$, which is not trivial, $\nabla_{xx}^2 L = \text{diag}(2, -2)$ is positive definite — the **second-order sufficient conditions hold** too.

KKT and the local duality gap (cont'd)

Yet $L(\cdot, \mu^*) = x_1^2 - x_2^2$ has a **saddle point** at x^* , so for every $\epsilon > 0$

$$\mathcal{L}_{\mathcal{B}(x^*, \epsilon)}(\mu^*) = \min_{\|x\| \leq \epsilon} (x_1^2 - x_2^2) = -\epsilon^2 < 0 = f(x^*).$$

The gap does tend to 0 with ϵ , which makes the mistake tempting, but it vanishes on no neighbourhood. And the failure has nothing to do with the optimality of x^* , which is global: it comes from the behaviour of the Lagrangian itself.

A sufficient condition is $\nabla_{xx}^2 L \succ 0$ on the **whole** space, and not only on the critical cone: above, $\text{diag}(2, -2)$ is positive definite on the cone but indefinite on $\mathbb{R}^2$.

Restoring a local zero gap

The **augmented Lagrangian** is precisely the standard remedy:

$$L_\rho(x, \lambda, \mu) = L(x, \lambda, \mu) + \frac{\rho}{2} \|h(x)\|^2,$$

whose Hessian at x^* is $\nabla_{xx}^2 L + \rho \nabla h(x^*) \nabla h(x^*)^T$. Under LICQ and the second-order conditions, it becomes positive definite for ρ large enough, so x^* locally minimizes L_ρ and the local gap vanishes.

On the previous counterexample, $L_\rho(x, 0) = x_1^2 + (\rho/2 - 1)x_2^2$, of Hessian $\text{diag}(2, \rho - 2)$: the gap is positive for $\rho \leq 2$ and null as soon as $\rho > 2$.

This is the motivation for the augmented Lagrangian methods, and in stochastic programming for **progressive hedging**, where the penalty is placed on the nonanticipativity constraints.

Active set

Definition (Active set)

The active set $\mathcal{A}(x)$ of the optimization problem

$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & f(x) \\ \text{s.t.} \quad & g_i(x) \leq 0, \quad i \in \mathcal{I} \\ & h_i(x) = 0, \quad i \in \mathcal{E}, \end{aligned}$$

in a feasible point x is the index set of the equality constraints and the active inequality constraints at that point:

$$\mathcal{A}(x) = \mathcal{E} \cup \{i \in \mathcal{I} \mid g_i(x) = 0\}$$

LICQ

The most popular constraint qualification is the LICQ.

Definition (LICQ)

Given a feasible point x and its active set $\mathcal{A}(x)$, the linear independence constraint qualification (LICQ) holds at x if the gradients of the active constraints, $\{\nabla_x h_i(x), i \in \mathcal{E}\} \cup \{\nabla_x g_i(x), i \in \mathcal{A}(x) \cap \mathcal{I}\}$, are linearly independent.